Sahaj Bhadja

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Summary

New college grad (M.S. CS 2026) with strong **CPU/architecture** fundamentals, **formal verification**-oriented validation experience, and expert **Python/C++** scripting. Skilled in testplan execution, **assertion-style** property checking, simulation debug, coverage analysis, and cross-team collaboration on complex system correctness.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Operating Systems, Analysis of Algorithms, Computer Architecture, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Operating Systems, Data Structures, Object-Oriented Programming, Computer Networks

Experience

USC Dornsife **Los Angeles, CA**
Formal Verification Engineer *September 2025 – Present*
- Defined verification scope and **testplans** for platform integrations serving 500+ users, executing regression suites, property-style validation checks, and release-readiness coverage gates across complex multi-component data workflows.
- Debugged failure scenarios with structured logs and deterministic replay for 5K+ monthly jobs, articulating verification coverage and root causes to engineering partners while sustaining **99%+ correctness** outcomes.

Grad8 **Los Angeles, CA**
Verification Engineering Intern *June 2025 – August 2025*
- Executed integration **testplans** for systems handling **10K+ daily requests**, validating timing behavior, edge-case properties, and defect reproduction through scripted **Python** diagnostics with cross-functional design and QA collaborators.
- Accelerated failure triage by **60%** using abstraction-friendly test scenarios, automated checks, and clear coverage reporting while learning verification standards under senior engineer mentorship in fast-paced delivery cycles.

USC AD Labs **Los Angeles, CA**
Verification Engineering Assistant *January 2025 – May 2025*
- Built closed-loop **simulation** evaluation pipelines across 100K+ trajectory records, developing scenario libraries and reusable harnesses to verify behavioral properties, stability metrics, and failure modes in complex architecture-driven research systems.
- Collaborated with researchers to document verification evidence, compare bounded proof-style rollout outcomes, and recommend abstraction strategies that improved planning stability by **15%** in simulation environments.

Aasma Technology Solutions **India**
Verification Automation Engineer - (SecureSync) *August 2023 – July 2024*
- Maintained **20+ automated verification suites** in Jenkins flows for distributed services, improving regression coverage by **68%** through assertion-like checks, mock environments, and fault-injection scenarios across Windows/Linux systems.
- Contributed **Python** flow scripts and integration tests that reduced production defects, supported hardware/software boundary debugging, and improved team efficiency through repeatable verification automation and structured failure artifacts.

MedDevice Test Systems **India**
Formal Verification Intern *January 2023 – April 2023*
- Implemented **Python** verification libraries for device communication scenarios, encoding lifecycle properties (pairing, reconnect, drop/recover) with deterministic logs, simulators, and bounded timing validation suitable for regulated correctness workflows.
- Integrated property-based test suites into CI pipelines with standardized coverage reports, enabling cross-team review of verification completeness and accelerating debug of intermittent failure conditions in connected device environments.

Skills

Verification: formal verification methodologies, assertion-based checking, temporal logic concepts, testplan development, coverage analysis, abstraction techniques, simulation validation, failure debug, bounded proofs
Languages & HDL: **Python**, **C++**, Java, TCL (familiar), Perl (familiar), **SystemVerilog** (familiar), Verilog (familiar), RTL debug concepts, scripting automation
Architecture & Tools: **CPU architecture**, GPU architecture (familiar), operating systems, algorithms, computer architecture, JasperGold (familiar), VC-Formal (familiar), Jenkins, Git, PyTest, analytical problem solving, distributed collaboration

Projects

CPU Scheduling Simulator Live | *C++, CPU Architecture, Simulation*
- Implemented preemptive **CPU scheduling** algorithms with interactive simulation controls and metric visualization across 1K+ process combinations, strengthening architecture understanding through correctness checks and edge-case behavioral analysis.

FormalTrace Suite | *Python, Property Assertions, Simulation*
- Developed a property-checking framework with environment assumptions, assertion rules, and coverage reporting over simulated state transitions, applying abstraction techniques to obtain bounded verification outcomes on complex finite-state behaviors.

ArchSim Validator | *Python, C++, Closed-Loop Verification*
- Built reusable validation modules for simulation rollouts, timing checks, and scenario regression comparisons, supporting debug of failure trajectories and articulate coverage summaries for architecture-sensitive autonomous-system test environments.